SQL OPERATIONS ON Dataset -1

Creation of tables(departments ,employees ,projects):

```
create table departments(
 department_id  int primary key,
 department_name varchar(50) not null);

 CREATE TABLE employees (
    employee_id int PRIMARY KEY,
    employee_name VARCHAR(50) NOT NULL,
    salary float(10,2),
    department_id int,
    manager_id int,
    FOREIGN KEY (department_id)
        REFERENCES departments(department_id),
    FOREIGN KEY (manager_id)
        REFERENCES employees(employee_id)
);
alter table employees modify salary double(10,2);
 select * from employees;
 CREATE TABLE projects (
    project_id int PRIMARY KEY,
    project_name varchar(100),
    employee_id int,
    status varchar(20),
    FOREIGN KEY (employee_id)
        REFERENCES employees(employee_id)
```

Inserting Values into tables:

```
insert into departments values(1,'IT');
INSERT INTO departments VALUES (2, 'HR');
INSERT INTO departments VALUES (3, 'Finance');
INSERT INTO departments VALUES (4, 'Marketing');

INSERT INTO employees VALUES (101, 'Rahul', 80000, 1, NULL);
INSERT INTO employees VALUES (102, 'Priya', 65000, 1, 101);
INSERT INTO employees VALUES (103, 'Kiran', 60000, 3, 101);
INSERT INTO employees VALUES (104, 'Anudeep', 45000, 1, 102);
INSERT INTO employees VALUES (105, 'Sneha', 50000, 2, 102);
INSERT INTO employees VALUES (106, 'Arjun', 40000, 3, 103);
INSERT INTO employees VALUES (107, 'Meena', 55000, 4, 103);
INSERT INTO employees VALUES (108, 'Vikram', 35000, 1, 102);

INSERT INTO projects VALUES (201, 'Cloud Migration', 101, 'Completed');
INSERT INTO projects VALUES (202, 'Web Application', 104, 'In Progress');
INSERT INTO projects VALUES (203, 'Payroll System', 103, 'Completed');
INSERT INTO projects VALUES (204, 'Recruitment Portal', 105, 'In Progress');
INSERT INTO projects VALUES (205, 'Network Monitoring', 102, 'Pending');
INSERT INTO projects VALUES (206, 'Marketing Analytics', 107, 'Completed');

 commit;
```

select * from employees;

employee_id	employee_name	salary	department_id	manager_id
101	Rahul	80000.00	1	NULL
102	Priya	65000.00	1	101
103	Kiran	60000.00	3	101
104	Anudeep	45000.00	1	102
105	Sneha	50000.00	2	102
106	Arjun	40000.00	3	103
107	Meena	55000.00	4	103

employees 17

select * from departments;

department_id	department_name
1	IT
2	HR
3	Finance
4	Marketing
NULL	NULL

select * from projects;

project_id	project_name	employee_id	status
201	Cloud Migration	101	Completed
202	Web Application	104	In Progress
203	Payroll System	103	Completed
204	Recruitment Portal	105	In Progress
205	Network Monitoring	102	Pending
206	Marketing Analytics	107	Completed
NULL	NULL	NULL	NULL

1)

```
/*finding employees who are earning more than 45000*/
select employee_id,employee_name,salary from employees where salary>45000;
```

OUTPUT:

employee_id	employee_name	salary
101	Rahul	80000.00
102	Priya	65000.00
103	Kiran	60000.00
105	Sneha	50000.00
107	Meena	55000.00
NULL	NULL	NULL

2)

```
 /*Displaying the name of the employee along with the department*/
SELECT e.employee_name, d.department_name
FROM employees e
INNER JOIN departments d
    ON e.department_id = d.department_id;
```

OUTPUT:

employee_name	department_name
Rahul	IT
Priya	IT
Anudeep	IT
Vikram	IT
Sneha	HR
Kiran	Finance
Arjun	Finance

3)

```
/*Aggregation with Group By and Having*/
select d.department_name,
count(e.employee_id) as total_employees,
round(avg(e.salary),2) as average_salary
from departments d
JOIN employees e
on d.department_id=e.department_id
group by d.department_name
having avg(e.salary)>45000;
```

OUTPUT:

department_name	total_employees	average_salary
IT	4	56250
HR	1	50000
Finance	2	50000
Marketing	1	55000

4)

```
/*Display the five highest-paid employees*/
select employee_name as emp_name,salary as emp_salary
from employees
order by salary desc
limit 5;
```

OUTPUT:

emp_name	emp_salary
Rahul	80000.00
Priya	65000.00
Kiran	60000.00
Meena	55000.00
Sneha	50000.00

5)

```sql
/*Finding the employees who are earning more than the average salary*/
select employee_name,salary from employees
where salary>(select avg(salary) from employees);
```

employee_name	salary
Rahul	80000.00
Priya	65000.00
Kiran	60000.00
Meena	55000.00

6)

```sql
/*Find employees who have not been assigned any project.*/
select e.employee_id,e.employee_name
from employees e
Left join projects p
on e.employee_id=p.employee_id
where p.project_id is null;
```

OUTPUT:

employee_id	employee_name
106	Arjun
108	Vikram

7)

```sql
/*Display each employee's name alongside their manager's name.*/
SELECT e.employee_name AS employee,
       m.employee_name AS manager
FROM employees e
LEFT JOIN employees m
    ON e.manager_id = m.employee_id
ORDER BY manager, employee;
```

OUTPUT:

	employee	manager
▶	Rahul	NULL
	Arjun	Kiran
	Meena	Kiran
	Anudeep	Priya
	Sneha	Priya
	Vikram	Priya
	Kiran	Rahul